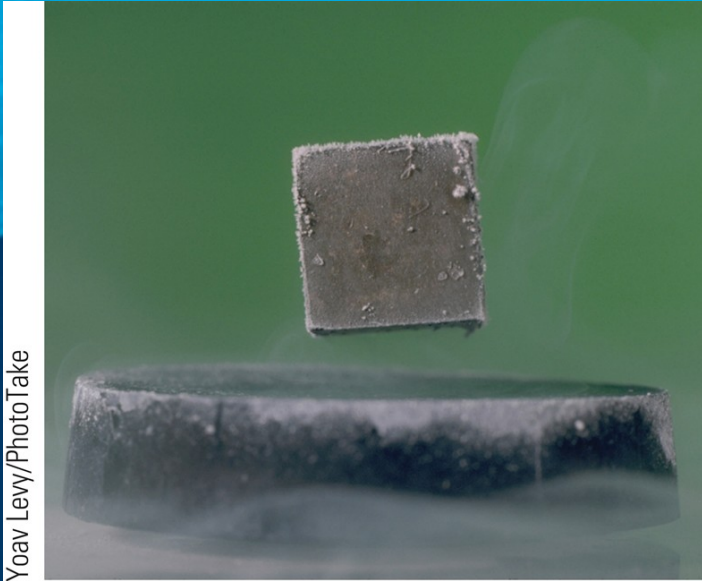




Yoav Levy/PhotoTake

Topic 17

Current and Resistance

Reading Question 17-1 (1 of 2)

Electrons flow through a circuit in the clockwise direction. The current is traveling

1. clockwise.
2. counterclockwise.
3. not at all.
4. in a direction impossible to tell without knowing what the positive charges are doing.

Reading Question 17-1 (2 of 2)

Electrons flow through a circuit in the clockwise direction. The current is traveling

1. clockwise.
2. **Answer: counterclockwise.**
3. not at all.
4. in a direction impossible to tell without knowing what the positive charges are doing.

Reading Question 17-2 (1 of 2)

The potential difference across a resistor R , and the current running through it, both increase. We know that resistance

1. decreases.
2. increases.
3. remains the same.
4. changes in ways we cannot determine with the information given.

Reading Question 17-2 (2 of 2)

The potential difference across a resistor R , and the current running through it, both increase. We know that resistance

1. decreases.
2. increases.
3. remains the same.
4. **Answer: changes in ways we cannot determine with the information given.**

Reading Question 17-3 (1 of 2)

A resistor, R , is made out of a particular material with resistivity ρ . The cylindrically shaped resistor has length L and radius r . To get four times as much resistance in your resistor you could

1. make the radius four times as big.
2. make the radius half as big.
3. make the radius one quarter the size.
4. make the radius twice as big.

Reading Question 17-3 (2 of 2)

A resistor, R , is made out of a particular material with resistivity ρ . The cylindrically shaped resistor has length L and radius r . To get four times as much resistance in your resistor you could

1. make the radius four times as big.
- 2. Answer: make the radius half as big.**
3. make the radius one quarter the size.
4. make the radius twice as big.

Reading Question 17-4 (1 of 2)

Which of the following best describes the inner workings of conductors?

1. Electrons collide at random, experiencing a net change of position equal to zero, except when exposed to an external electric field.
2. The electrons start at one electrode and travel directly to the other electrode, stopping there.
3. The electrons' drift speed depends only on the material of the conductor and not on the voltage difference applied to it.
4. None of these choices is correct.

Reading Question 17-4 (2 of 2)

Which of the following best describes the inner workings of conductors?

- 1. Answer: Electrons collide at random, experiencing a net change of position equal to zero, except when exposed to an external electric field.**
2. The electrons start at one electrode and travel directly to the other electrode, stopping there.
3. The electrons' drift speed depends only on the material of the conductor and not on the voltage difference applied to it.
4. None of these choices is correct.

Reading Question 17-5 (1 of 2)

The battery powering a light bulb is disconnected. As the light bulb's heat dissipates, how will the resistance and current change?

1. The resistance will decrease and the current will increase.
2. The resistance will increase and the current will be zero.
3. The resistance will increase and the current will decrease.
4. The resistance will decrease and the current will be zero.

Reading Question 17-5 (2 of 2)

The battery powering a light bulb is disconnected. As the light bulb's heat dissipates, how will the resistance and current change?

1. The resistance will decrease and the current will increase.
2. The resistance will increase and the current will be zero.
3. The resistance will increase and the current will decrease.
4. **Answer: The resistance will decrease and the current will be zero.**

Reading Question 17-6 (1 of 2)

Which of the following best describes superconductors?

1. conductors that have less than half the resistivity of iron
2. conductors with essentially no resistance below a critical temperature, peculiar to each material
3. conductors which have a drop in their resistivity as their temperatures are lowered
4. Conductors are materials which are very useful in a variety of ways; hence the name "superconductor."

Reading Question 17-6 (2 of 2)

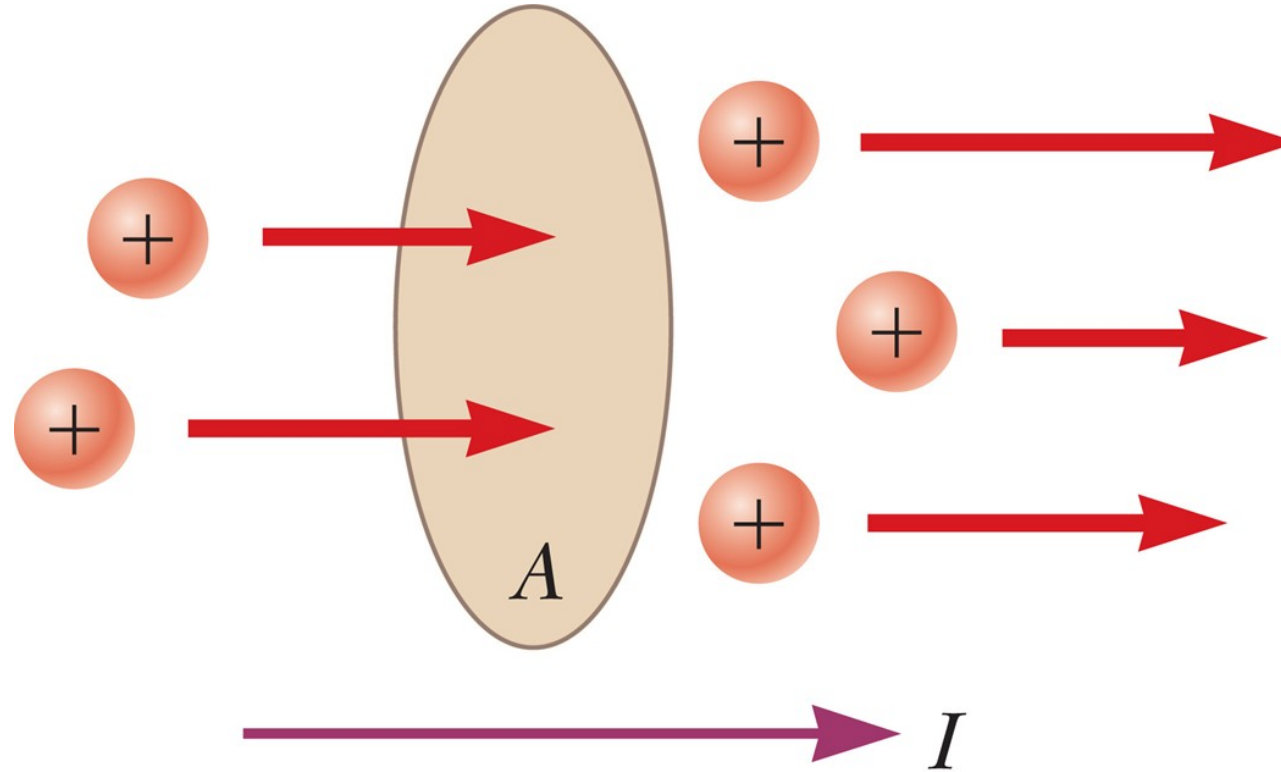
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4. Conductors are materials which are very useful in a variety of ways; hence the name "superconductor."

Topic 17: Current and Resistance

TOPIC DISCUSSION

Electric Current (1 of 4)

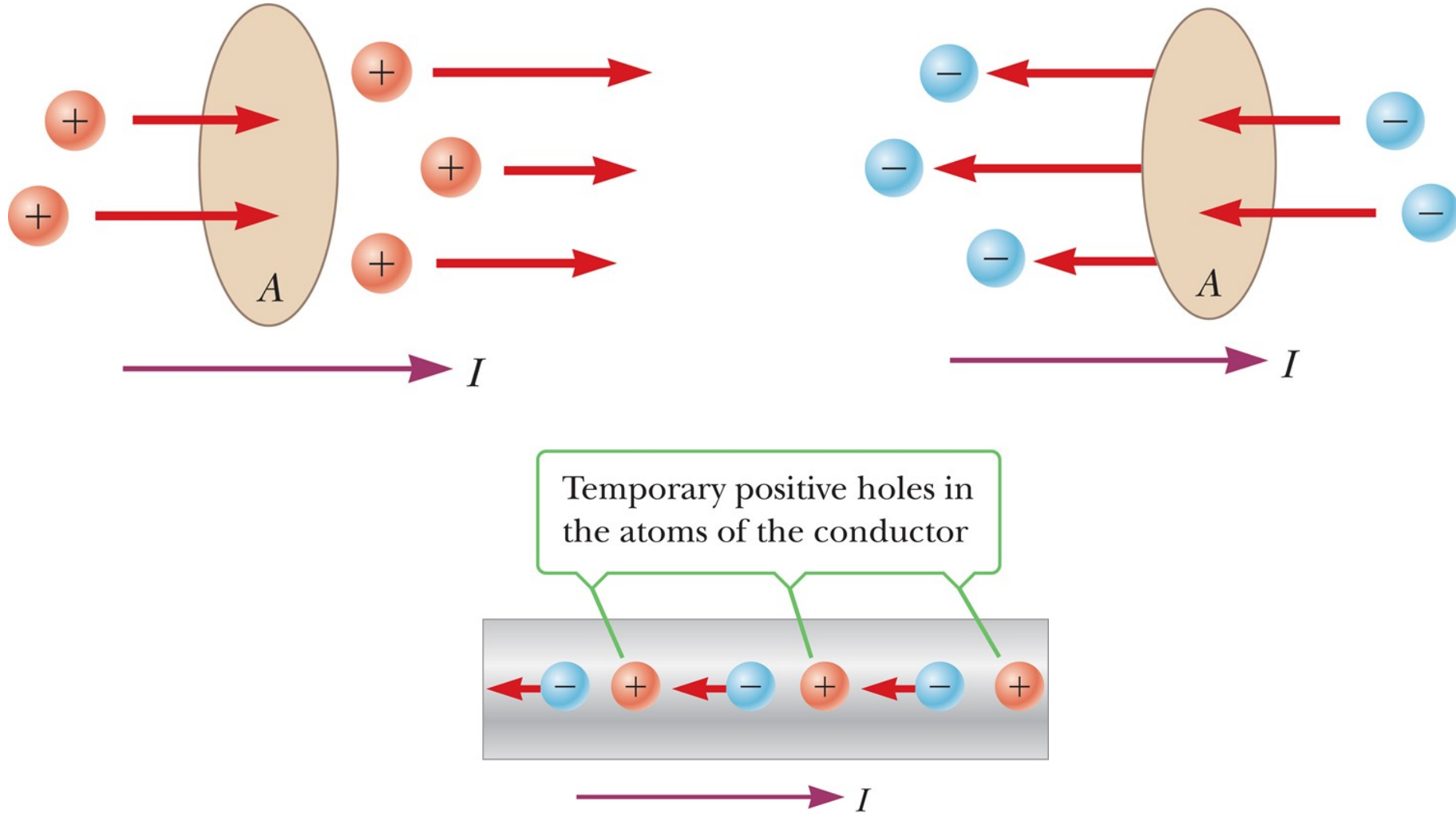


$$I_{\text{av}} \equiv \frac{\Delta Q}{\Delta t} \quad \text{SI unit: C/s or A}$$

Electric Current (2 of 4)

$$I = \lim_{\Delta t \rightarrow 0} I_{\text{av}} = \lim_{\Delta t \rightarrow 0} \frac{\Delta Q}{\Delta t}$$

Electric Current (3 of 4)



Electric Current (4 of 4)

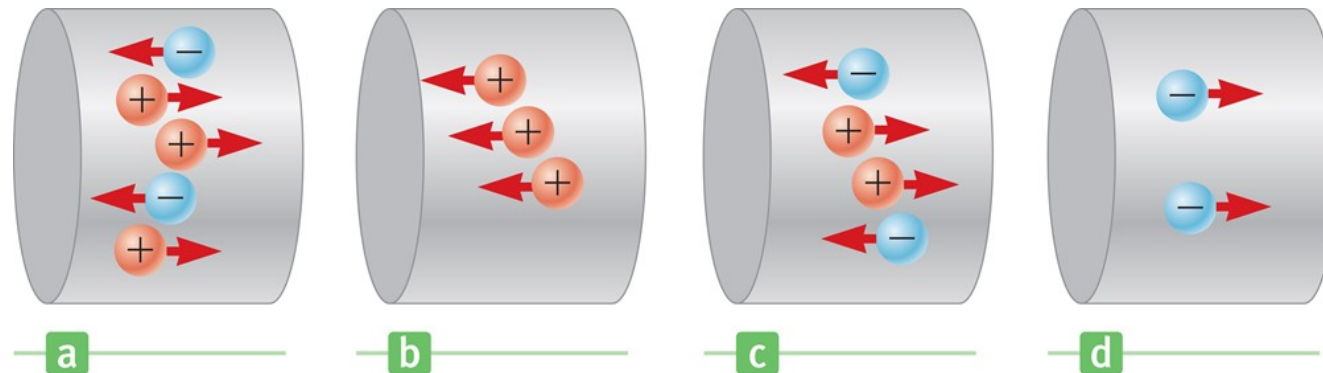
$$\Delta U_{\text{charges}} = q \Delta V$$

$$\Delta U_{\text{wire}} = -q \Delta V$$

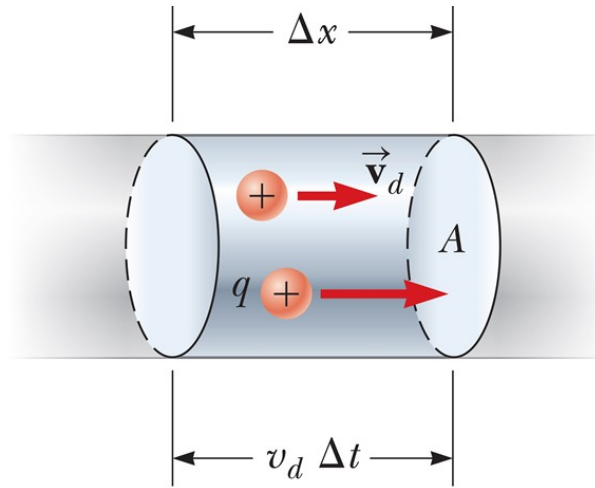
Think – Pair – Share 1

Consider positive and negative charges moving horizontally through the four regions in the figure below. Rank the currents in these four regions, from lowest to highest. (I_a is the current in Figure (a), I_b the current in Figure (b), etc.)

1. I_d, I_a, I_c, I_b
2. I_a, I_c, I_b, I_d
3. I_c, I_a, I_d, I_b
4. I_d, I_b, I_c, I_a
5. I_a, I_b, I_c, I_d



A Microscopic View: Current and Drift Speed (1 of 2)



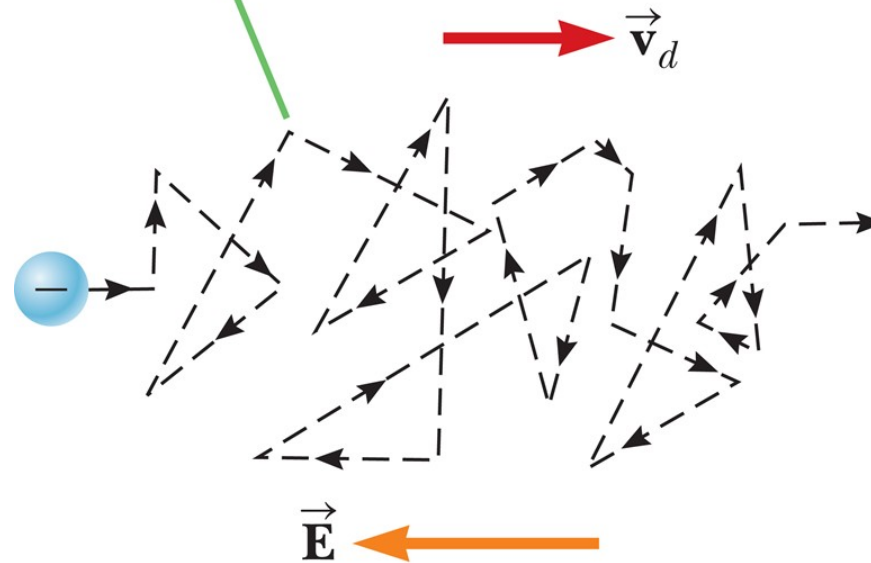
$$\Delta Q = \text{number of carriers} \times \text{charge per carrier} = (nA \Delta x)q$$

$$\Delta x = v_d \Delta t \rightarrow \Delta Q = (nAv_d \Delta t)q$$

$$I = \lim_{\Delta t \rightarrow 0} \frac{\Delta Q}{\Delta t} = nqv_d A$$

A Microscopic View: Current and Drift Speed (2 of 2)

Although electrons move with average velocity $\vec{v}_d$, collisions with atoms cause sharp, momentary changes of direction.

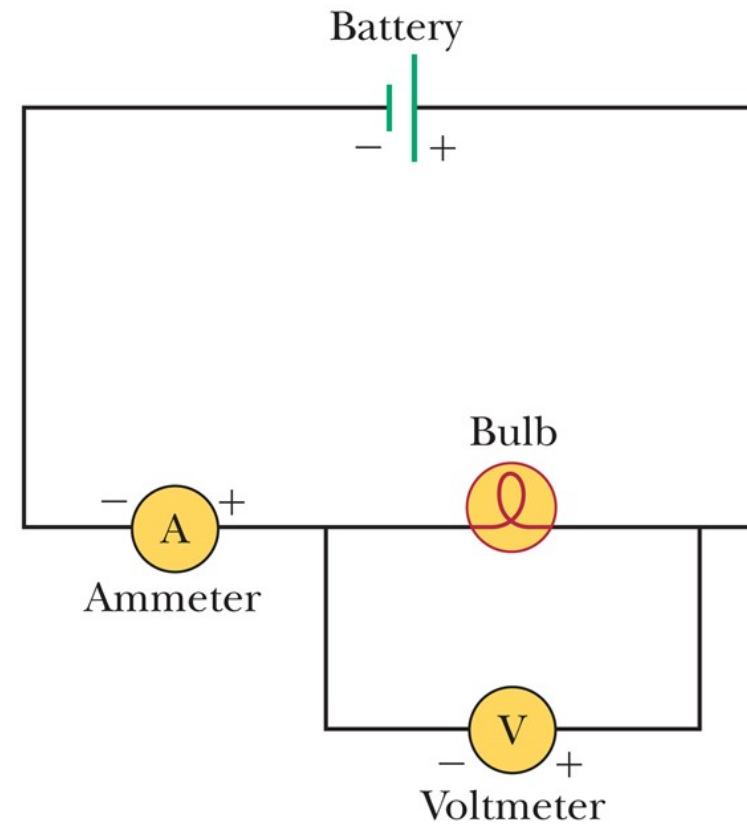
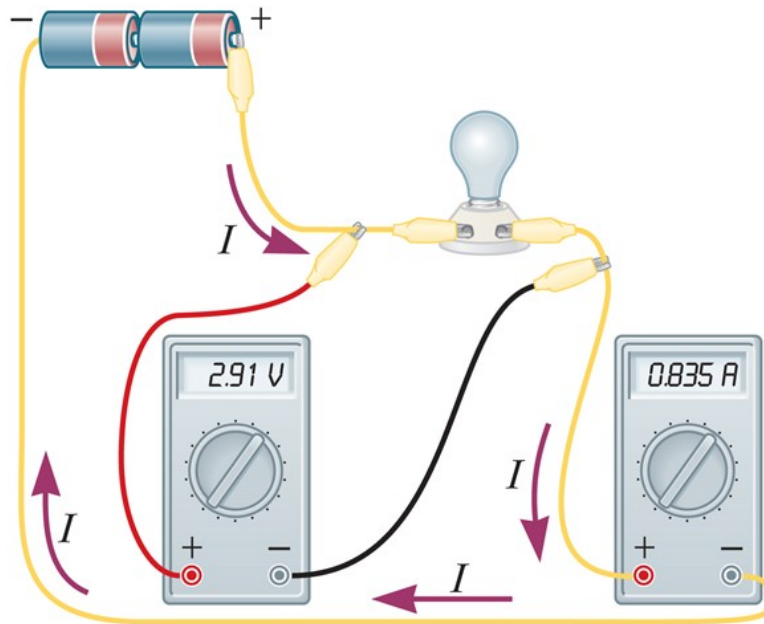


Think – Pair – Share 2

Suppose a current-carrying wire has a cross-sectional area that gradually becomes smaller along the wire so that the wire has the shape of a very long, truncated cone. How does the drift speed vary along the wire?

1. It slows down as the cross section becomes smaller.
2. It speeds up as the cross section becomes smaller.
3. It doesn't change.
4. More information is needed.

Current and Voltage Measurements in Circuits (1 of 2)



Current and Voltage Measurements in Circuits (2 of 2)

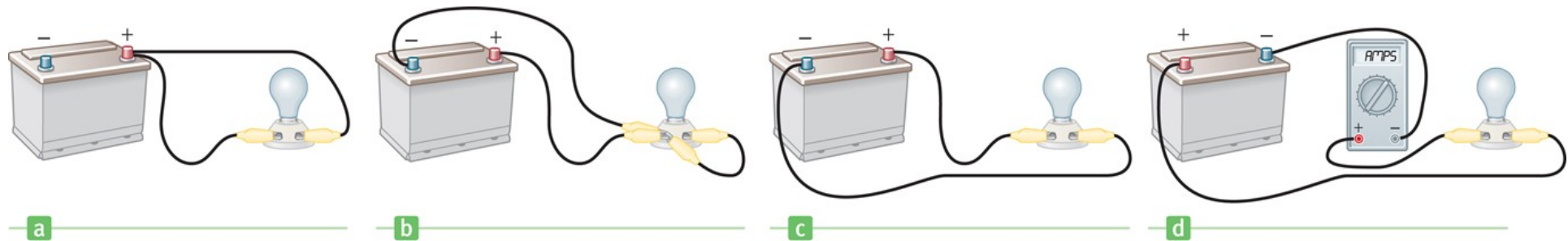


iStockphoto/Yury Kosourov

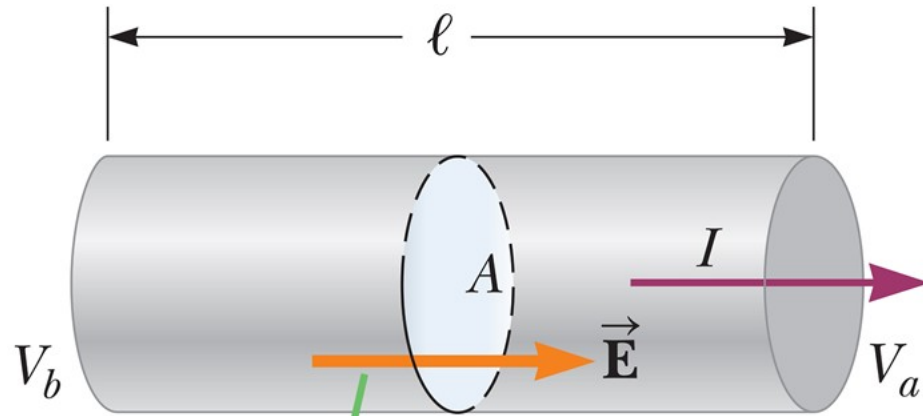
Think – Pair – Share 3

Look at the four “circuits” shown in the figure below, and select those that will light the bulb.

1. a and c
2. b only
3. c and d
4. All will light the bulb.



Resistance and Ohm's Law (1 of 3)



The potential difference $\Delta V = V_b - V_a$ creates the electric field $\vec{E}$ that produces the current I .

$$\Delta V = IR$$

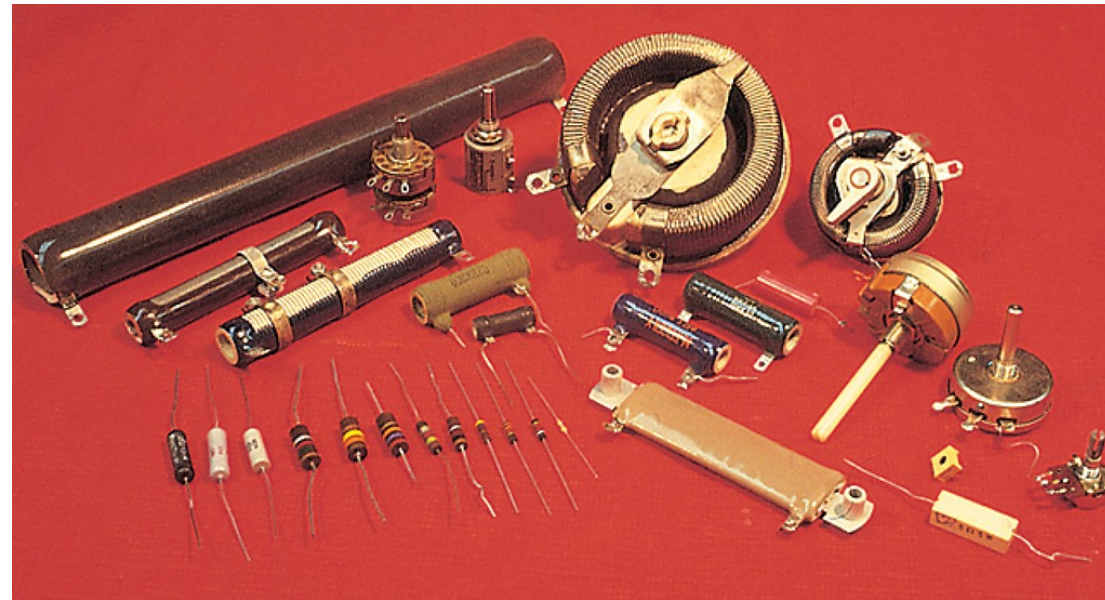
$$R \equiv \frac{\Delta V}{I} \quad \text{SI unit: V/A} = \Omega$$

Example:

$$R = \frac{120 \text{ V}}{6 \text{ A}} = 20 \Omega$$

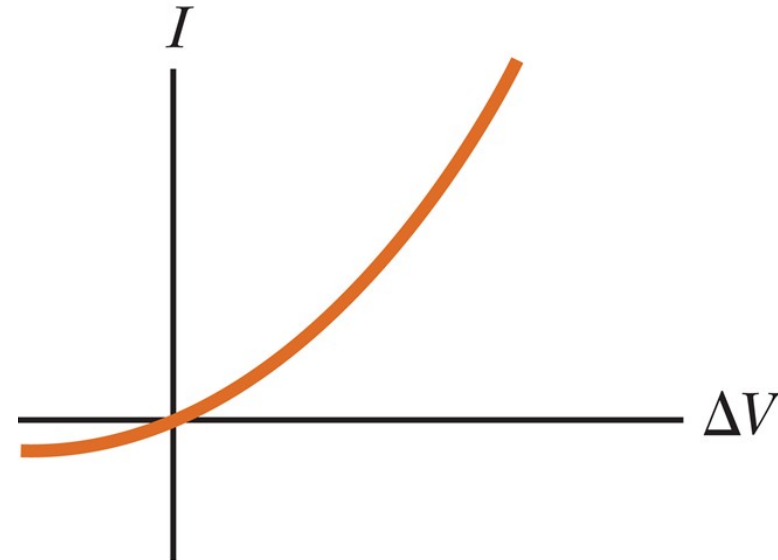
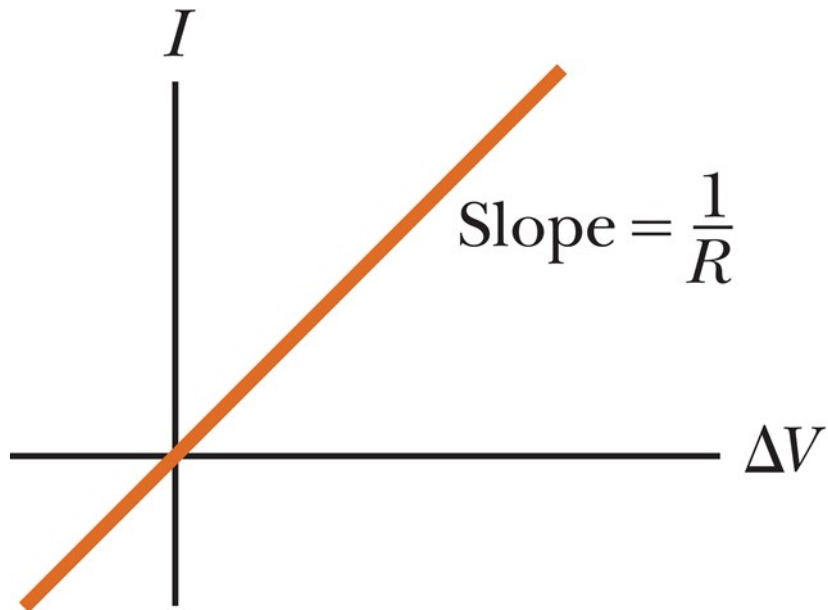
Resistance and Ohm's Law (2 of 3)

$$\Delta V = IR \quad (\text{Ohm's law})$$



Courtesy of Henry Leap and
Jim Lehman

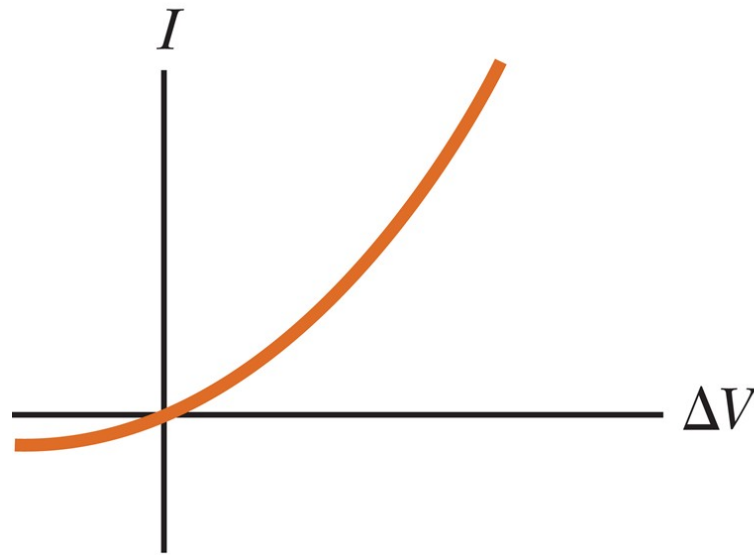
Resistance and Ohm's Law (3 of 3)



Think – Pair – Share 4

In the figure below, does the resistance of the diode increase or decrease as the positive voltage ΔV increases?

1. increase
2. decrease

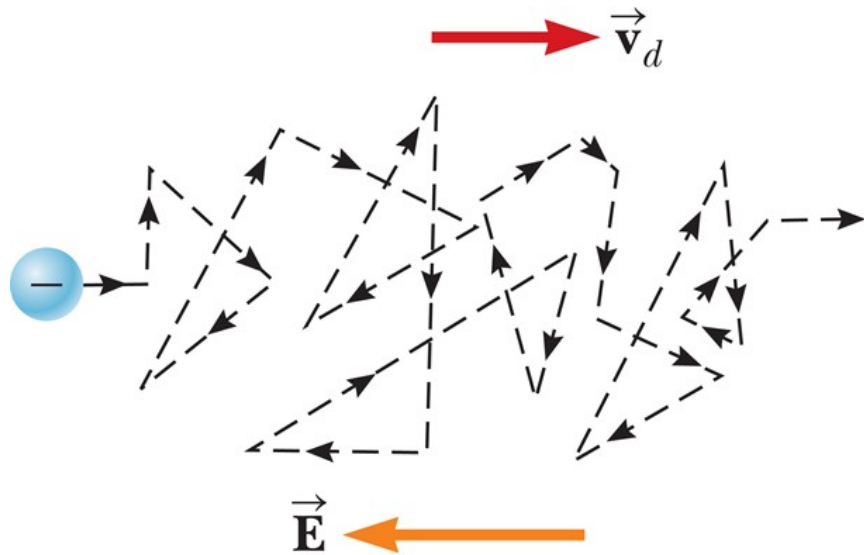


Think – Pair – Share 5

All electric devices are required to have identifying plates that specify their electrical characteristics. The plate on a certain steam iron states that the iron carries a current of 6.00 A when connected to a source of $1.20 \times 10^2\text{ V}$. What is the resistance of the steam iron?

1. $0.0500\ \Omega$
2. $20.0\ \Omega$
3. $36.0\ \Omega$

Resistivity



$$R = \rho \frac{\ell}{A}$$

Table 17.1 Resistivities and Temperature Coefficients of Resistivity for Various Materials (at 20°C)

Material	Resistivity ($\Omega \cdot \text{m}$)	Temperature Coefficient of Resistivity [($^{\circ}\text{C}$) $^{-1}$]
Silver	1.59×10^{-8}	3.8×10^{-3}
Copper	1.7×10^{-8}	3.9×10^{-3}
Gold	2.44×10^{-8}	3.4×10^{-3}
Aluminum	2.82×10^{-8}	3.9×10^{-3}
Tungsten	5.6×10^{-8}	4.5×10^{-3}
Iron	10.0×10^{-8}	5.0×10^{-3}
Platinum	11×10^{-8}	3.92×10^{-3}
Lead	22×10^{-8}	3.9×10^{-3}
Nichrome ^a	150×10^{-8}	0.4×10^{-3}
Carbon	3.5×10^{-5}	-0.5×10^{-3}
Germanium	0.46	-48×10^{-3}
Silicon	640	-75×10^{-3}
Glass	10^{10} – 10^{14}	
Hard rubber	$\approx 10^{13}$	
Sulfur	10^{15}	
Quartz (fused)	75×10^{16}	

^aA nickel-chromium alloy commonly used in heating elements.

Think – Pair – Share 6

Suppose an electrical wire is replaced with one having every linear dimension doubled (i.e., the length and radius have twice their original values). The wire now has

1. more resistance than before.
2. less resistance.
3. The same resistance.

Temperature Variation of Resistance (1 of 2)



Lawrence Manning/Corbis

$$\rho = \rho_0 [1 + \alpha(T - T_0)]$$

$$R = \rho \frac{\ell}{A}$$

$$R = R_0 [1 + \alpha(T - T_0)]$$

Temperature Variation of Resistance (2 of 2)

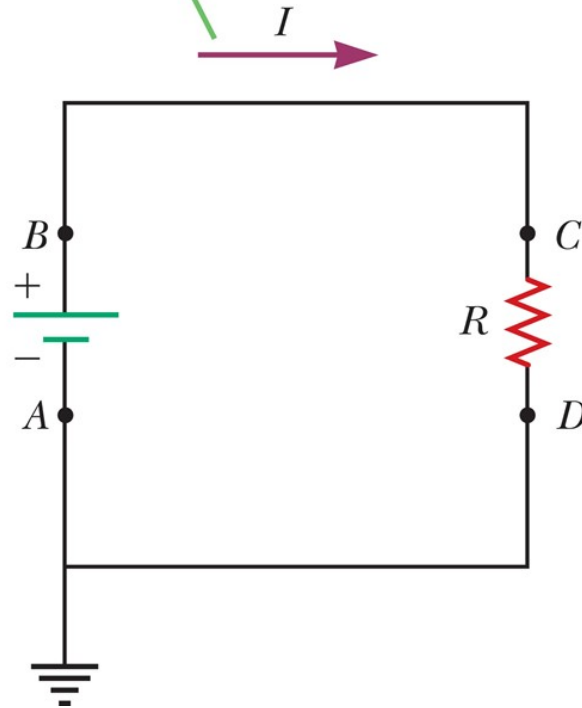
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Quartz (fused)	75×10^{16}	

^aA nickel-chromium alloy commonly used in heating elements.

Electrical Energy and Power (1 of 2)

Positive current travels clockwise from the positive to the negative terminal of the battery.



Electrical Energy and Power (2 of 2)

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta Q}{\Delta t} \Delta V = I \Delta V$$

$$P = I \Delta V$$

$$\Delta V = IR \quad \rightarrow \quad P = I^2 R = \frac{(\Delta V)^2}{R}$$

$$1 \text{ kWh} = (10^3 \text{ W})(3600 \text{ s}) = 3.60 \times 10^6 \text{ J}$$

Think – Pair – Share 7

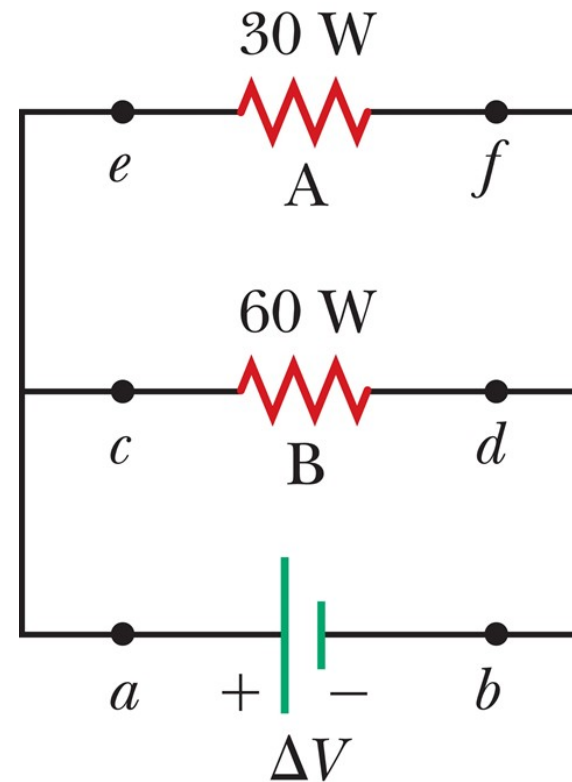
A voltage ΔV is applied across the ends of a Nichrome heater wire having a cross-sectional area A and length L . The same voltage is applied across the ends of a second Nichrome heater wire having a cross-sectional area A and length $2L$. Which wire gets hotter?

1. The shorter wire does.
2. The longer wire does.
3. More information is needed.

Think – Pair – Share 8

For the two resistors shown in the figure below, rank the currents at points *a* through *f* from largest to smallest.

1. $I_a = I_b > I_e = I_f > I_c = I_d$
2. $I_a = I_b > I_c = I_d > I_e = I_f$
3. $I_e = I_f > I_c = I_d > I_a = I_b$



Think – Pair – Share 9

Two resistors, A and B, are connected in a series circuit with a battery. The resistance of A is twice that of B. Which resistor dissipates more power?

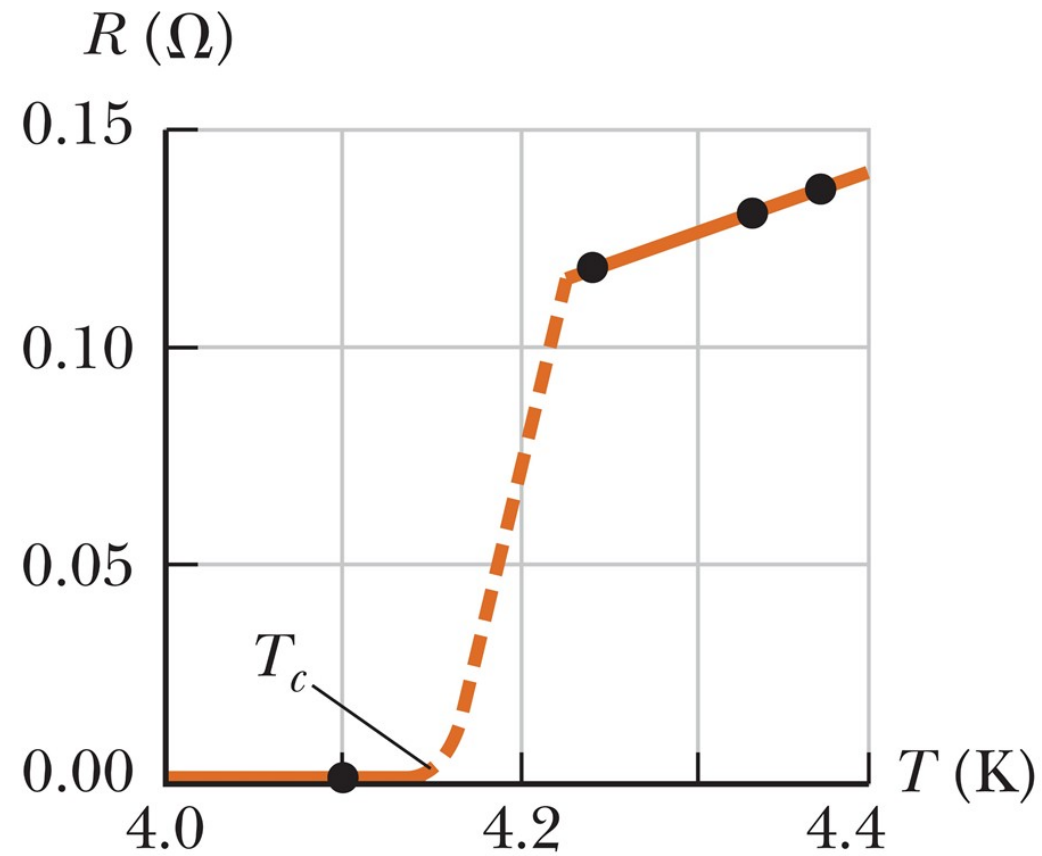
1. Resistor A does.
2. Resistor B does.
3. More information is needed.

Think – Pair – Share 10

The diameter of wire A is greater than the diameter of wire B, but their lengths and resistivities are identical. For a given voltage difference across the ends, what is the relationship between P_A and P_B , the dissipated power for wires A and B, respectively?

1. $P_A = P_B$
2. $P_A < P_B$
3. $P_A > P_B$

Superconductors (1 of 3)

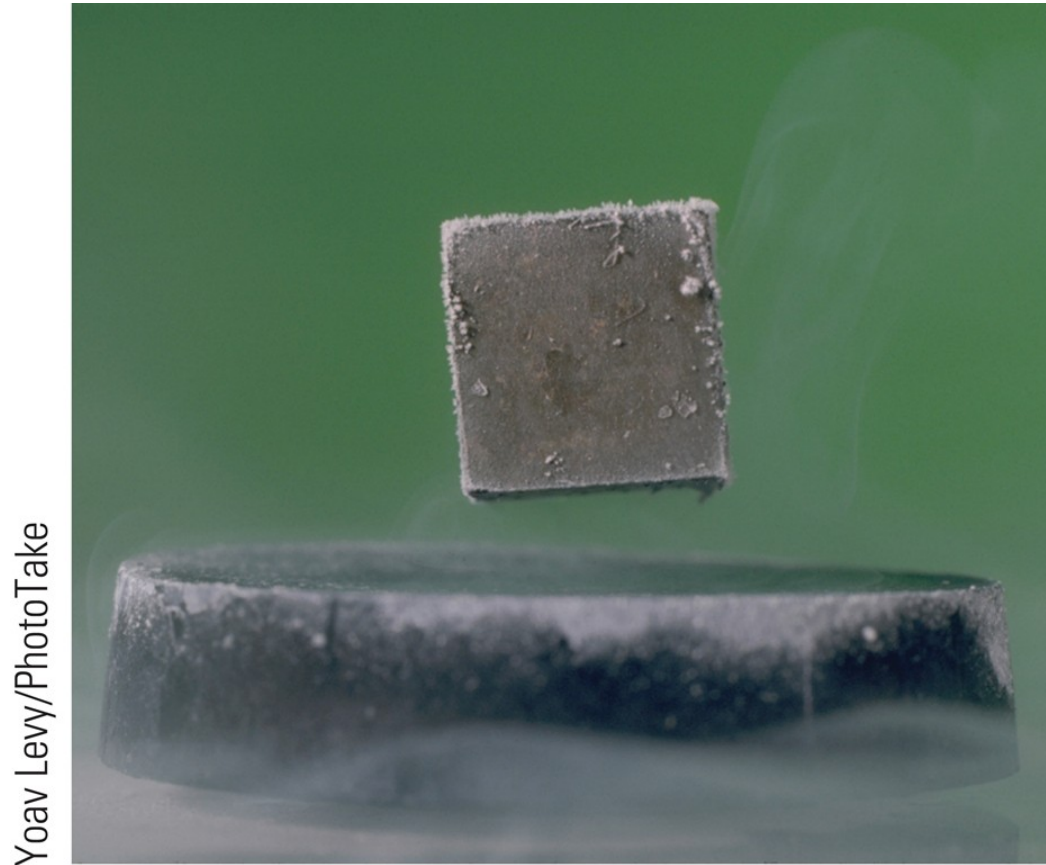


Superconductors (2 of 3)

Table 17.2 Critical Temperatures for Various Superconductors

Material	$T_c(\text{K})$
Zn	0.88
Al	1.19
Sn	3.72
Hg	4.15
Pb	7.18
Nb	9.46
Nb ₃ Sn	18.05
Nb ₃ Ge	23.2
YBa ₂ Cu ₃ O ₇	92
Bi–Sr–Ca–Cu–O	105
Tl–Ba–Ca–Cu–O	125
HgBa ₂ Ca ₂ Cu ₃ O ₈	134

Superconductors (3 of 3)

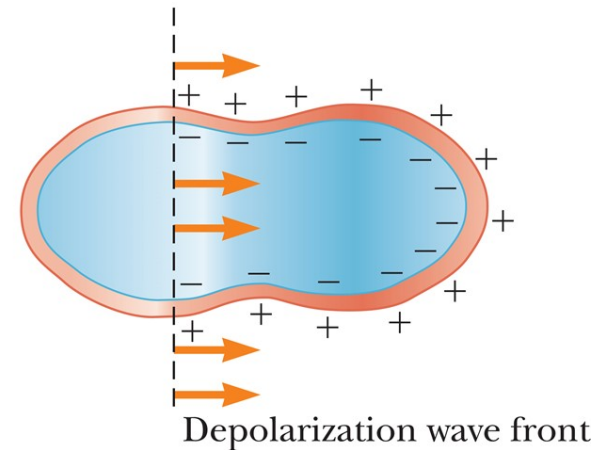
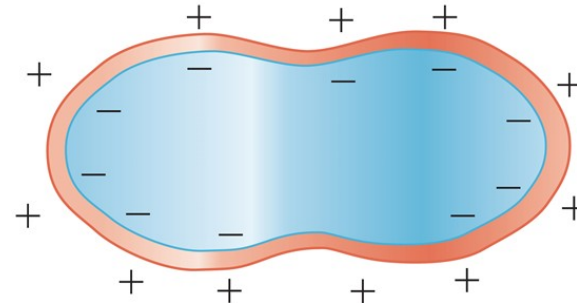
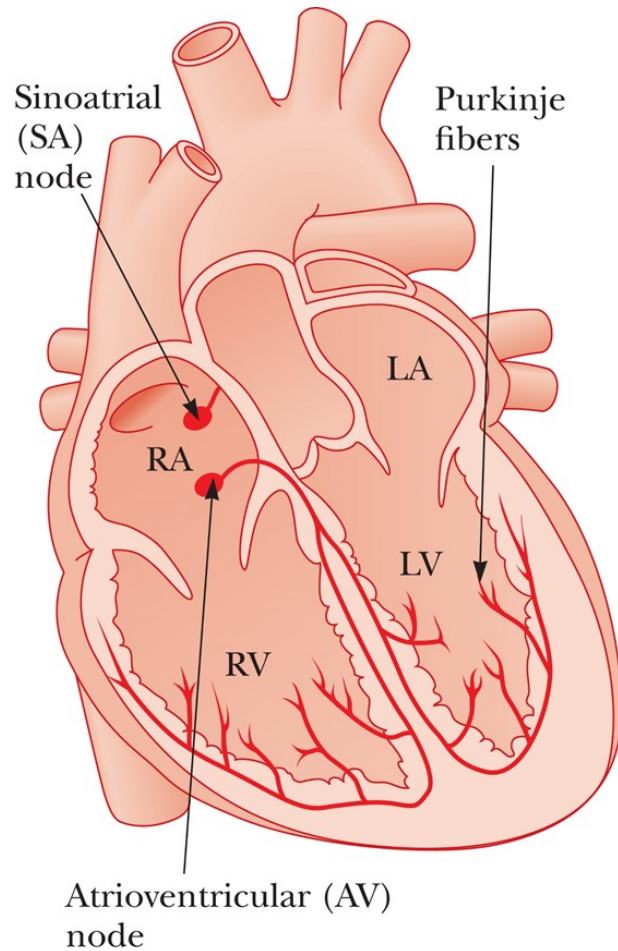


Yoav Levy/PhotoTake

Electrical Activity in the Heart: Electrocardiograms (1 of 5)

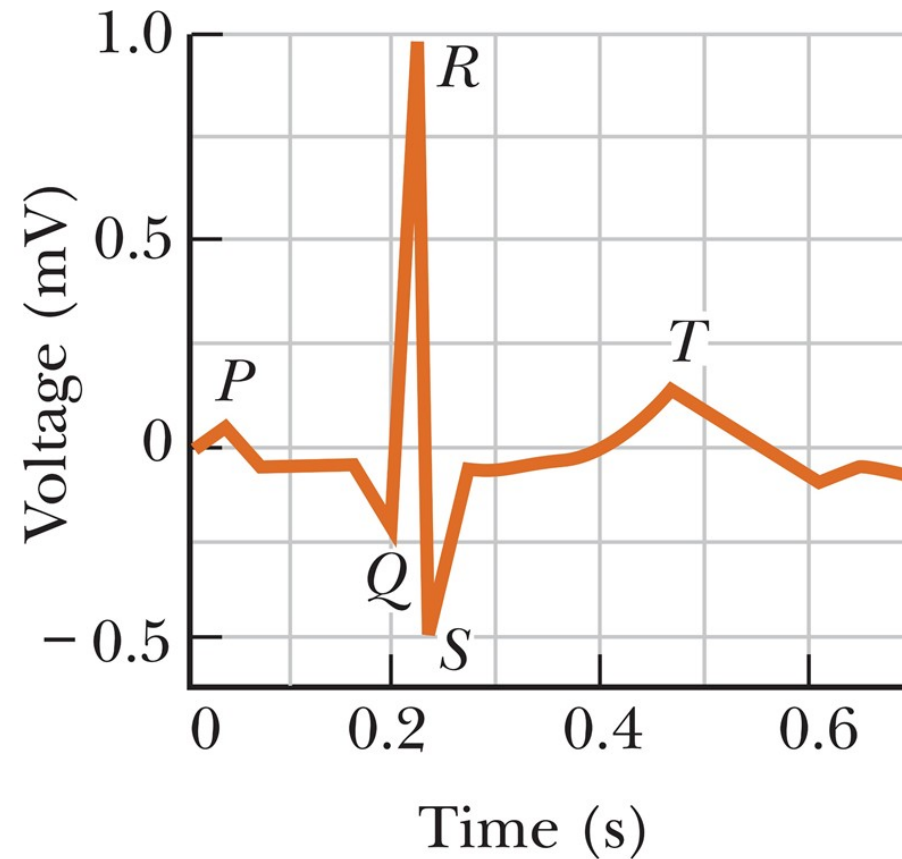


Electrical Activity in the Heart: Electrocardiograms (2 of 5)

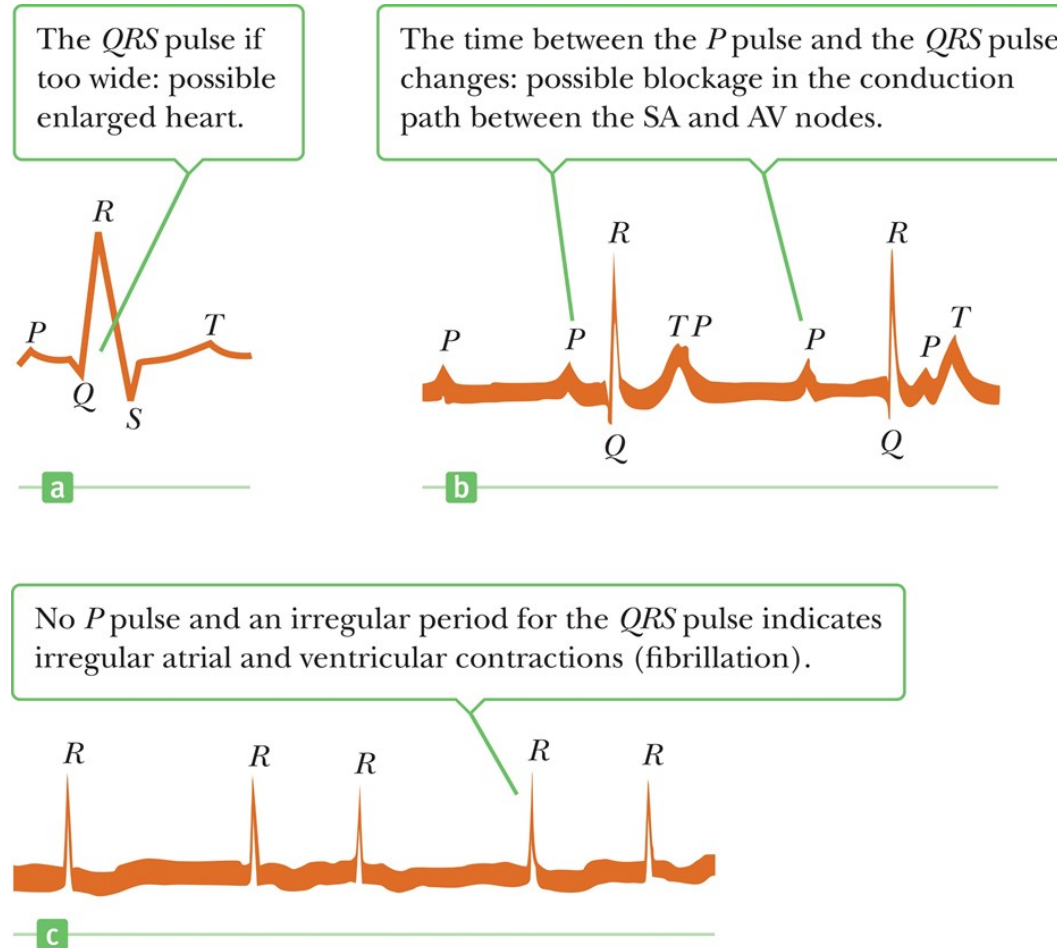


Electrical Activity in the Heart:

Electrocardiograms (3 of 5)



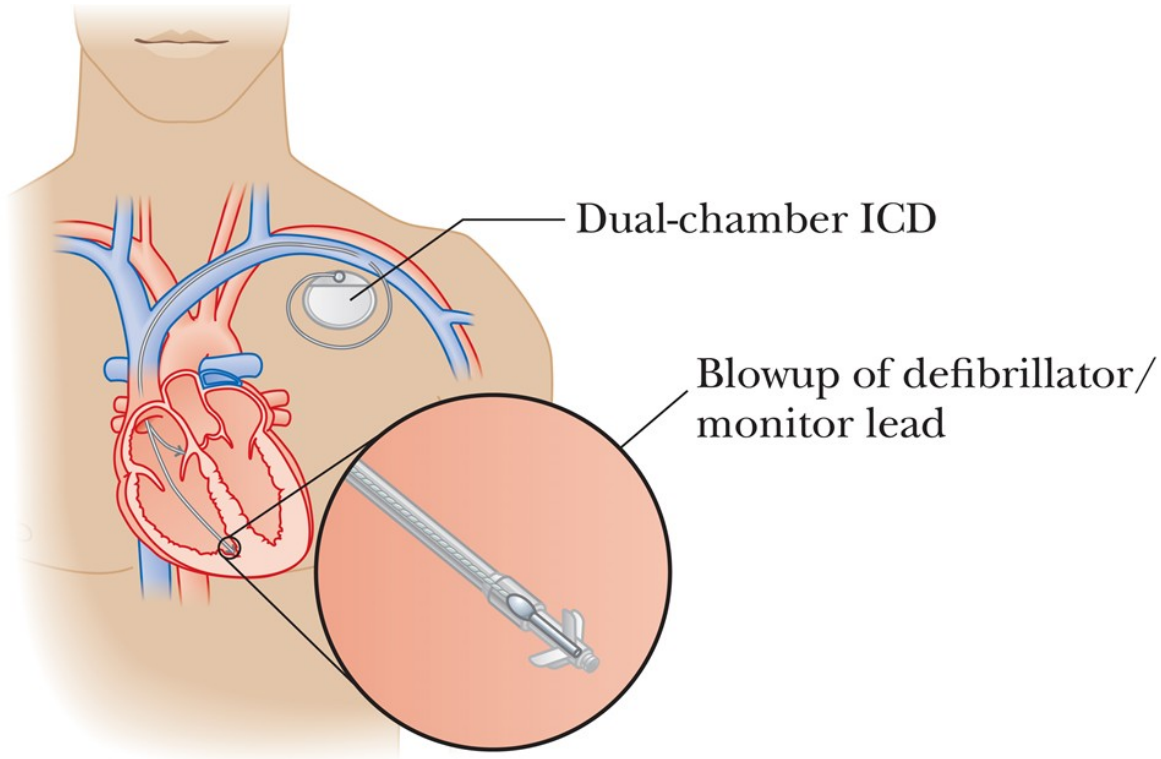
Electrical Activity in the Heart: Electrocardiograms (4 of 5)



Electrical Activity in the Heart: Electrocardiograms (5 of 5)



An Emergency Room in Your Chest (1 of 4)



Courtesy of Medtronic, Inc.

b

An Emergency Room in Your Chest (2 of 4)

1. Monitoring
2. Storing signals
3. Easy to reprogram
4. Signal analysis and comparison
5. Supplying pacing signals or a high-voltage pulse
6. Automatically adjusting the number of pacing pulses per minute

An Emergency Room in Your Chest (3 of 4)

Table 17.3 Typical Properties of Implanted Cardioverter Defibrillators

Physical Specifications

Mass (g)	85
Size (cm)	$7.3 \times 6.2 \times 1.3$ (about five stacked silver dollars)
Antitachycardia Pacing	ICD delivers a burst of critically timed low-energy pulses
Number of bursts	1–15
Burst cycle length (ms)	200–552
Number of pulses per burst	2–20
Pulse amplitude (V)	7.5 or 10
Pulse width (ms)	1.0 or 1.9

An Emergency Room in Your Chest (4 of 4)

Table 17.3 Typical Properties of Implanted Cardioverter Defibrillators

Physical Specifications

High-Voltage Defibrillation

Pulse energy (J) 37 stored/33 delivered

Pulse amplitude (V) 801

Bradycardia Pacing

A dual-chamber ICD can steadily deliver repetitive pulses to both the atrium and the ventricle

Base frequency
(beats/minute) 40–100

Pulse amplitude (V) 0.25–7.5

Pulse width (ms) 0.05, 0.1–1.5, 1.9